# An Introduction to Ergodicity Economics

Ole Peters and Alexander Adamou

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<u>Behavioral protocol:</u> choose process with max $g$, outperform in the long run.

# $v$-and-$x$ correspondence

Role of $v(x)$: linearizes $x$ in time.

Can find $x$ as $x = v^{(-1)}(\gamma t)$.

# Stochastic growth

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Then find $x(t)$ as before as $x = v^{(-1)}(\gamma t + \sigma dW)$

Measure speed of $x(t)$ like before:

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<u>Note:</u> exact correspondence to deterministic case.

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> (Trivial: utility function $u(x)$ is extra degree of freedom.)

1940s: axiomatization – max $\mathbb{E}\left[u(x(t+\Delta t))\right]$ interpreted as consistent idiosyncratic preferences.

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Multiply with constant $\Delta t$, same behavior:
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add constant $v(x(t))$, same behavior:
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$\rightarrow$ expected-utility theory, except one letter

$$\mathbb{E}[v(x(t+\Delta t))] = \mathbb{E}[u(x(t+\Delta t))].$$

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Non-linear "utility:" people maximize wealth.

Properly, consistently, over time.

Non-ergodicity $\rightarrow$ different from max $\mathbb{E}\left[x\right]$.

No (immediate) evidence of idiosyncratic risk preferences, aversion, seeking, bias.

Different behaviors:

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